

10x Cost, 10x Capability?

A Model Selection Framework for Requirement Quality Tagging in B2B SaaS

Master Glossary

Technical, Product, Financial & Governance Terms of Reference

Compiled as a shared reference across the project's five domains -- Product, Technical, Financial, Risk & Governance, and Operating Cadence -- so every term used in the PRD, harness, memo, and decision log means the same thing to everyone reading it.

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1. Technical & AI / LLM Terms

Core vocabulary for how large language models work, how they are called via API, and how their outputs are automatically scored.

LLM (Large Language Model) — A neural network trained on massive text corpora to predict and generate human-like text. *Project context: The core technology being benchmarked in this project.*

API (Application Programming Interface) — A defined way for one piece of software to request services from another. *Project context: The interface used to send prompts to each model and receive responses.*

Token — The basic unit of text an LLM processes -- roughly three-quarters of a word in English. *Project context: Pricing and context limits are measured in tokens, not words or characters.*

Tokenization — The process of splitting raw text into tokens before it is fed into a model.

Context Window — The maximum number of tokens (input + output combined) a model can handle in one call. *Project context: Exceeding it truncates or rejects the request.*

Prompt — The instruction/input text sent to a model. *Project context: Held identical across all four models so the comparison stays fair.*

System Prompt — A separate instruction block that sets a model's role or behaviour, distinct from the user's message. *Project context: Used to define the rubric's tagging instructions consistently for every model.*

Prompt Engineering — The practice of crafting prompt wording and structure to reliably get the desired output format and quality.

Temperature — A sampling parameter controlling randomness in a model's output; lower values are more deterministic. *Project context: Usually set low (near 0) for benchmarking, to reduce noise between repeated runs.*

Top-p / Top-k Sampling — Alternative randomness controls that restrict which tokens a model may sample at each step.

Inference — The act of running a trained model to produce an output, as opposed to training it. *Project context: Every benchmark call in this project is one inference request.*

Latency — The time between sending a request and receiving the full response. *Project context: Logged per call as one of the three trade-off dimensions (with cost and quality).*

Throughput — How many requests or tokens a system can process per unit of time. *Project context: Relevant if a chosen model cannot keep up with real production volume.*

Frontier Model — Industry shorthand for the most capable, usually most expensive model tier available at a given time.

Mid-Tier Model — A model priced and positioned between the frontier and cheap tiers.

Cheap / Fast Model — A smaller, cheaper, lower-latency model, often with reduced reasoning capability.

Open-Source Model — A model whose weights are publicly released and can be self-hosted or run via a free-tier host. *Project context: One of the four roster slots in this project's model lineup.*

Model Weights — The learned numerical parameters of a trained model. *Project context: "Open-source" specifically refers to these being publicly downloadable.*

Fine-Tuning — Further training a pretrained model on a narrower dataset to specialise its behaviour. *Project context: Out of scope for this project, which relies on prompting only.*

Zero-Shot Prompting — Asking a model to perform a task with no worked examples given in the prompt.

Few-Shot Prompting — Providing a small number of examples inside the prompt to guide the model's output format or behaviour.

Chain-of-Thought (CoT) — A prompting technique where the model is asked to reason step-by-step before giving its final answer. *Project context: Often used inside a judge prompt to make its scoring more reliable.*

LLM-as-Judge — Using a (typically strong) LLM to automatically score another model's output against a rubric, instead of exhaustive human review. *Project context: The evaluation method at the heart of this project's Part 2 / MVP.*

Rubric — An explicit, written set of scoring criteria/dimensions used consistently by both the human and the judge.

Ground Truth / Gold Label — The accepted "correct" answer or label for a data point, established by domain experts. *Project context: GitReq supplies these; they anchor the calibration step.*

Human-in-the-Loop (HITL) — A workflow where a human reviews, corrects, or validates AI outputs at some stage of the pipeline. *Project context: This project's manual calibration pass is a HITL step.*

Calibration (of a judge) — Checking an automated judge's scores against human or gold judgments to establish how much it can be trusted.

Inter-Rater Agreement — A statistical measure of how consistently two raters (e.g. human vs. judge) assign the same scores or labels.

Cohen's Kappa — A statistic for inter-rater agreement that corrects for the agreement expected by pure chance. *Project context: More rigorous than a raw percentage-agreement figure.*

Confusion Matrix — A table showing predicted vs. actual labels across all classes. *Project context: Reveals exactly which ISO 25010 categories the judge confuses.*

Precision — Of all items labelled as a given class, the fraction that were actually correct.

Recall — Of all items that truly belong to a class, the fraction the model/judge correctly identified.

F1 Score — The harmonic mean of precision and recall -- a single balanced accuracy-style metric.

Accuracy — The fraction of all predictions that were correct. *Project context: Simplest metric, but can mislead on imbalanced classes.*

Hallucination — When an LLM generates plausible-sounding but false or fabricated content. *Project context: A key risk factor to flag, especially for the cheaper model tier.*

Bias (Judge Bias) — A systematic skew in a judge's scoring -- e.g. favouring longer answers (verbosity bias) or the first option shown (position bias). *Project context: Called out explicitly in the Risk & Governance domain.*

Structured Output / JSON Schema — Requiring a model's response to follow a fixed, machine-parseable format so it can be programmatically scored and stored.

API Rate Limit — A cap on how many requests or tokens can be sent to a provider within a given time window. *Project context: Relevant when running a full benchmark sweep across the dataset.*

SDK (Software Development Kit) — A provider's official code library that simplifies calling their API, compared to raw HTTP requests.

SQL (Structured Query Language) — The standard language for querying relational databases. *Project context: Used to query the benchmark results table.*

SQLite — A lightweight, file-based relational database requiring no separate server. *Project context: The suggested store for this project's `input x model -> score, cost, latency` table.*

Pandas / DataFrame — A Python library and its core tabular data structure, used for data manipulation and analysis.

Matplotlib — A Python charting library. *Project context: Used to produce the cost-vs-quality efficiency frontier chart.*

Git — A version control system that tracks changes to code and files over time.

GitHub — A cloud platform for hosting Git repositories. *Project context: Also the original source of the GitReq dataset's issues.*

Repository (Repo) — A project's full set of tracked files and version history.

Commit — A saved snapshot of changes in a Git repository.

Branch — A parallel, independent line of development within a repo.

Pull Request (PR) — A proposed set of changes submitted for review before merging into the main codebase.

Unit Test — Automated code that checks a small piece of functionality behaves as expected. *Project context: Used to validate the prompting and judge harnesses.*

Schema Validation — Automatically checking that a piece of data (e.g. a judge's JSON output) matches an expected structure before it is trusted.

CI/CD (Continuous Integration / Continuous Deployment) — Automated pipelines that test, and optionally deploy, code on every change. *Project context: An optional polish item for this project's technical deliverables.*

README — The top-level documentation file explaining what a repository does and how to reproduce it.

Streamlit — A Python framework for quickly building interactive web dashboards. *Project context: An optional stretch deliverable for exploring benchmark results.*

NLP (Natural Language Processing) — The broader AI field focused on understanding and generating human language.

Embedding — A numerical vector representation of text that captures semantic meaning. *Project context: Foundational LLM concept, not core to this project's scope.*

ISO/IEC 25010 — An international standard defining software product quality characteristics: functionality, performance efficiency, usability, reliability, security, maintainability, portability, and compatibility. *Project context: The backbone of this project's rubric, via the GitReq dataset's labels.*

Functional Requirement (FR) — A requirement describing what a system should do -- a specific behaviour or feature.

Non-Functional Requirement (NFR) — A requirement describing how a system should behave (performance, security, usability, etc.) rather than a specific feature.

SRS (Software Requirements Specification) — A formal document describing a system's intended functional and non-functional requirements. *Project context: The format underlying the PROMISE_exp dataset.*

INVEST — A mnemonic for good agile user stories: Independent, Negotiable, Valuable, Estimable, Small, Testable.

GitReq — This project's primary dataset: real GitHub issues expert-labelled against ISO/IEC 25010 quality categories. *Project context: Locked as the project's dataset of record.*

PROMISE_exp — A classic academic dataset of labelled software requirements (functional vs. non-functional). *Project context: Used only as an optional, secondary cross-check, not the core dataset.*

2. Data & Statistics Terms

Statistical vocabulary needed to describe the dataset, report latency, and defend the judge-calibration numbers.

Sample Size (n) — The number of data points or examples used in an analysis. *Project context: This project's dataset is intentionally small (25-40 items) but curated for realism.*

Mean / Average — The sum of a set of values divided by their count.

Median — The middle value in a sorted dataset. *Project context: More robust to outliers than the mean -- useful for latency reporting.*

Standard Deviation — A measure of how spread out values are around the mean.

Percentile (e.g. p50, p95, p99) — A value below which a given percentage of observations fall. *Project context: "p95 latency" means 95% of requests were faster than that value.*

Distribution — The overall pattern or shape of how values are spread across a dataset.

Outlier — A data point that differs significantly from the rest of the dataset. *Project context: Can skew average cost/latency figures if not flagged.*

Correlation — A statistical measure of how two variables move together. *Project context: E.g. does higher cost correlate with higher quality score?*

Agreement Rate — The simple percentage of times two raters (human and judge) assign the same score or label. *Project context: The headline number from the manual calibration step.*

3. Product & PM Terms

The product-management vocabulary used to scope, phase, and communicate the project as a real decision system, not just a script.

PRD (Product Requirements Document) — A document defining the problem, goals, scope, and requirements for a product or feature before building it.

Acceptance Criteria — Specific, checkable conditions that must be true for a deliverable to be considered done. *Project context: Defined for each of this project's three phases.*

Success Metrics / KPI (Key Performance Indicator) — Quantifiable measures used to judge whether a goal has been achieved. *Project context: E.g. judge-human agreement %, cost delta, latency SLA.*

MVP (Minimum Viable Product) — The smallest version of a product or system that delivers real, usable value. *Project context: Phase 2 of this project's roadmap.*

0 to 1 (Zero to One) — Shorthand for the moment something goes from not existing to existing in a working form for the first time. *Project context: Marks the MVP milestone: the first populated results table.*

Scope — The defined boundary of what is, and is not, included in a project. *Project context: "Scope creep" is the risk of it silently expanding -- e.g. "full execution plan" ballooning past a benchmark + memo.*

Stakeholder — Any person or group with an interest in, or influence over, a project's outcome. *Project context: The fictional B2B SaaS client in this project's narrative.*

Persona — A fictional but realistic representation of a user or customer type, used to ground decisions.

User Story — A short, structured feature description from an end-user's perspective, often "As a [user], I want [goal], so that [reason]."

Decision Memo — A short document presenting a recommendation backed by evidence, written for a decision-maker audience. *Project context: This project's headline deliverable, written after seeing the results.*

Decision Log — A running record of key project decisions and the reasoning behind them. *Project context: Captures why GitReq, this rubric, and this model roster were chosen.*

Roadmap — A high-level, phased plan showing what will be built and in what order. *Project context: The Discovery -> MVP -> Launch structure of this project.*

Trade-off Matrix — A table comparing options across multiple weighted criteria to support a decision.

Build vs. Buy — The classic decision of whether to build a custom solution or use/purchase an existing one. *Project context: Analogous to this project's "switch model or keep it" framing.*

Feature Flag — A toggle that lets a team turn a feature on or off (or for a subset of users) without deploying new code. *Project context: Relevant to a gradual model-switch rollout in the deployment strategy.*

Canary Release / Canary Rollout — Releasing a change to a small subset of traffic or users first, to catch problems before a full rollout.

Rollback — Reverting to a previous, known-good state after a release causes problems. *Project context: Part of the deployment strategy's rollback criteria.*

SLA (Service Level Agreement) — A commitment to a measurable service standard, such as latency or uptime.

Backlog — The full list of pending work items not yet scheduled into a specific phase.

Sprint — A fixed, short time-box (commonly one to two weeks) in Agile development for completing a set of work.

Definition of Done (DoD) — A team's shared checklist for when a piece of work is truly complete. *Project context: Formalised here as each phase's Acceptance Criteria.*

Go-to-Market (GTM) — The plan for how a product or feature reaches and is adopted by its intended users.

Change Management — The structured approach to helping stakeholders or users adapt to a change. *Project context: Relevant if a production model swap requires team retraining or workflow updates.*

Risk Register — A documented list of identified risks, their likelihood/impact, and mitigation plans.

4. Financial & Economic Terms

The economic vocabulary that turns raw API pricing into a business case a finance stakeholder can act on.

TCO (Total Cost of Ownership) — The full projected cost of using a solution over a period or volume, not just its per-call sticker price. *Project context: Extrapolates per-call cost to the client's real monthly request volume.*

Unit Economics — The direct revenues and costs associated with a single unit. *Project context: Here, a single API call or a single tagged requirement.*

Cost per Call / Cost per 1K Tokens — The price of a single API request, or of processing 1,000 tokens. *Project context: The basic building block of the entire cost analysis.*

Break-Even Analysis — Calculating the point (e.g. volume or time) at which a more expensive option's benefits start to outweigh its extra cost.

ROI (Return on Investment) — The ratio of benefit gained to the cost of achieving it, often expressed as a percentage.

Sensitivity Analysis — Testing how much a conclusion or recommendation changes if a key assumption (e.g. a model's price) changes. *Project context: Checks whether the memo's recommendation is a knife-edge result.*

Efficiency Frontier / Pareto Frontier — The set of options not "dominated" by any other option on both axes being compared. *Project context: Visualised as the project's core cost-vs-quality chart.*

Opportunity Cost — The value of the next-best alternative given up when a choice is made.

Sunk Cost — A cost already incurred that cannot be recovered, and so should not influence a forward-looking decision.

Marginal Cost — The cost of producing or processing one additional unit. *Project context: E.g. the cost of one more tagged requirement.*

CAPEX vs. OPEX — Capital Expenditure (upfront investment) vs. Operational Expenditure (ongoing running cost). *Project context: Pay-per-call LLM APIs are an OPEX model.*

Gross Margin — Revenue minus the direct cost of delivering a product or service, as a percentage of revenue. *Project context: Shows how model cost eats into a SaaS product's margin.*

COGS (Cost of Goods Sold) — The direct costs attributable to delivering a product or service. *Project context: For an AI feature, LLM API spend is often booked here.*

Vendor Lock-In — The risk or cost of depending on a single provider, making it hard or expensive to switch later. *Project context: Explicitly flagged as a Risk & Governance deliverable.*

Amortization — Spreading a fixed cost over time or usage volume. *Project context: E.g. the one-time cost of running this benchmark, spread across future decisions it informs.*

5. Risk & Governance (Responsible AI) Terms

Terms covering the safety, privacy, and accountability considerations that keep the project's recommendation honest.

Model Risk — The risk that a model's outputs are wrong, biased, or misused in a way that causes harm or bad decisions.

PII (Personally Identifiable Information) — Any data that could identify a specific individual. *Project context: A key concern when handling real business or requirement text.*

Data Anonymization / Scrubbing — Removing or masking identifying details from data before use or publication. *Project context: Relevant if any client-style example text needs cleaning before a public repo.*

Data Residency — Legal or contractual requirements about which geographic location data must be stored or processed in.

SOC 2 — An auditing standard/report assessing a vendor's data security controls. *Project context: Often required for enterprise B2B SaaS deals -- flagged as "N/A here, but would need review in production."*

GDPR (General Data Protection Regulation) — EU law governing personal data privacy and protection. *Project context: Relevant if any benchmark data could contain personal information.*

Explainability / Interpretability — The degree to which a model's decision-making process can be understood by a human.

Drift (Model Drift / Data Drift) — When a model's real-world performance degrades over time because incoming data or conditions no longer match what it was built or tested on. *Project context: A key trigger in the Operating Cadence monitoring plan.*

Compliance — Adhering to relevant laws, regulations, and internal policies.

Audit Trail — A recorded, chronological trace of actions and decisions that can be reviewed later to verify what happened and why. *Project context: Served here by the Decision Log.*

6. Operating Cadence & Monitoring Terms

Vocabulary for treating the benchmark as a living system with a lifecycle, rather than a one-time snapshot.

Monitoring Plan — A defined set of metrics and checks used to continuously observe a system's real-world performance after launch.

Re-benchmark Trigger — A predefined condition that should prompt re-running the benchmark. *Project context: E.g. a new model release, a >15% price change, or an SLA breach.*

Observability — The broader practice of understanding a system's internal state from its external outputs (logs, metrics, traces).

Alerting Threshold — A predefined limit that, when crossed, triggers a notification for human review.

Versioning — Tracking distinct, comparable versions of an artifact (dataset, rubric, model roster) so future re-runs can be measured against a known baseline.